



Question No.	Answer	Mark												
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1 (b) (i)	Video	J1												
1 (b) (ii)	Animation	J1												
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2 (a) (i)	Production Production is the design phase of a project to produce the content based on the idea.	J1 J2												
2 (a) (ii)	Post-production Post-production is the assembly, editing and testing the final product.	J1 J2												
2 (b) (i)	Multimedia elements and events are organized as objects in a structural framework or process.	J2												
2 (b) (ii)	Multimedia elements are presented and organized along a timeline.	J2												



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Question No.	Answer	Mark									
2 (c)	Hand gesture-based navigation refers to control and handling multimedia applications through hand gesture recognition scheme.	J2									
2 (d) (i)	Online	J1									
2 (d) (ii)	• Web based : YouTube Platform • Cloud based : Google Drive	Choose any one (1) J1									
2 (d) (iii)	• Slow Loading: Sometimes videos and other multimedia take a long time to load, making it frustrating for users with slower internet connections. • Compatibility Issues: Multimedia may not work on all devices or browsers, causing some users to miss out on the content. • Security Risks: There's a risk of content being stolen or misused, making it challenging to protect multimedia content from unauthorized use. • Copyright Problems: It's easier for people to copy and distribute multimedia content without permission, leading to copyright issues for creators. *Accept any suitable answer	Choose any one (1) J2									
3 (a)	<table border="1"> <thead> <tr> <th>Criteria</th><th>File processing system</th><th>Database system</th></tr> </thead> <tbody> <tr> <td>Data Redundancy</td><td>Data is often duplicated / High data redundancy. [J1]</td><td>Minimize data redundancy / Low data redundancy. [J1]</td></tr> <tr> <td>Data Sharing</td><td>Limited data sharing as data is stored in separate files. [J1]</td><td>Facilitates efficient data sharing among multiple users. [J1]</td></tr> </tbody> </table>	Criteria	File processing system	Database system	Data Redundancy	Data is often duplicated / High data redundancy. [J1]	Minimize data redundancy / Low data redundancy. [J1]	Data Sharing	Limited data sharing as data is stored in separate files. [J1]	Facilitates efficient data sharing among multiple users. [J1]	J1,J1 J1,J1
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3 (b)	• Database systems are complex, difficult, and time consuming to design. • Substantial hardware and software start-up cost.	Choose any one (1) J2									



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	• Damage to database affects virtually all applications program. • Extensive conversion costs in moving from a file-based system to a database system. • Initial training required for all programmers and users. *Accept any suitable answer	
4 (a)	• Data integrity refers to the correctness and consistency of stored data. • Data integrity refers to the accuracy, consistency, and reliability of data.	Choose any one (1) J2
4 (b) (i)	Form	J1
4 (b) (ii)	Report	J1
5 (a)	• GPS is a satellite navigation system used to determine the ground position of an object. • The Global Positioning System (GPS) relies on a network of satellites to provide geolocation and time information to users worldwide. The functioning of GPS involves the use of a network of satellites, ground stations, and receivers to determine accurate positioning data.	Choose any one (1) J2
5 (b)	• Waze • Google Maps • Garmin • Sygic • HERE WeGo *Accept any suitable answer	Choose any two (2) J1 J1
5 (c) (i)	Cellular Radio Transmission	J1
5 (c) (ii)	Infrared Transmission	J1
5 (c) (iii)	Microwave / Satellite Transmission	J1
5 (c) (iv)	Cellular Radio Transmission	J1
5 (c) (v)	Microwave / Satellite Transmission	J1
5 (d) (i)	Fiber Optic Cable	J1
5 (d) (ii)	Twisted Pair Cable	J1
5 (d) (iii)	Coaxial Cable	J1
5 (e) (i)	Modem	J1
5 (e) (ii)	Wireless Access Point	J1
5 (e) (iii)	Switch	J1
5 (e) (iv)	Network Card / Network Interface Card / Network Adapter Card	J1
5 (e) (v)	Router	J1
5 (e) (vi)	Repeater	J1
6 (a) (i)	Type – Subject Directory Example: • Google Directory	J1 Choose any one (1) J1



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Question No.	Answer	Mark
	• Yahoo! Directory	
6 (a) (ii)	Type – Search Engine Example: • Google search • Yahoo! search	J1 Choose any one (1) J1
6 (b)	Subject Directory - A subject directory classifies webpages in an organized set of categories, such as sports or shopping, and related subcategories. Search Engine - Software that finds websites, webpages, images, videos, news, maps, and other information related to a specific topic.	Choose any one (1) J2
6 (c) (i)	Using boolean operator	J1
6 (c) (ii)	Using phrases	J1
6 (c) (iii)	Using field search	J1
6 (d) (i)	Online entertainment	J1
6 (d) (ii)	e-Commerce	J1
6 (d) (iii)	Online News	J1
6 (d) (iv)	Social Networking / Social Media	J1
6 (d) (v)	Online Education	J1
6 (e) (i)	- Web browsing privacy encompasses what information about individuals is available, how it is used, and by whom. - Web browsing privacy refers to the protection of an individual's personal information and online activities while using the internet.	Choose any one (1) J2
6 (e) (ii)	• Treat email as a formal communication method • Don't use personal email platforms for business • Encrypt sensitive emails • Secure confidential attachments • Always enabled 2-step verification *Accept any suitable answer	Choose any two (2) J1 J1
6 (e) (iii)	• Emphasis on digital literacy and technological proficiency. • Integration of innovative teaching methods and technologies.	Choose any one (1) J2



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Question No.	Answer	Mark
	• Customized and adaptive learning experiences tailored to individual student needs. • Greater connectivity and collaboration on a global scale. • Focus on critical thinking, problem-solving, and creativity. • Introduction of virtual and augmented reality tools for immersive learning. • Shift towards interdisciplinary and STEM (Science, Technology, Engineering, Mathematics) education. • Increased demand for continuous skill development and lifelong learning. • Integration of data analytics for informed decision-making in education. • Preparation of students for the evolving job market and the skills required in the digital age. *Accept any suitable answer	

END OF ANSWER SCHEME